Performance Testing

Date	15 July 2026
Team ID	************
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Performance testing evaluates how the system behaves under expected and peak load conditions.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Development Server, Web Browser (Manual Functional Testing)
Type of Testing	Functional Testing & Basic Load Testing
Target Module / API	POST /submit (Loan Application → Data Preprocessing → Random Forest Prediction → Result Page)
Test Environment	Local Machine (Flask Development Server)
Test Date	13 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single applicant submits the loan application	1	5	Prediction returned in under 1 second
2	Multiple users submit applications simultaneously	10	30	All requests return correct predictions with no errors
3	Multiple submissions with different applicant details	20	60	Stable predictions and consistent response time

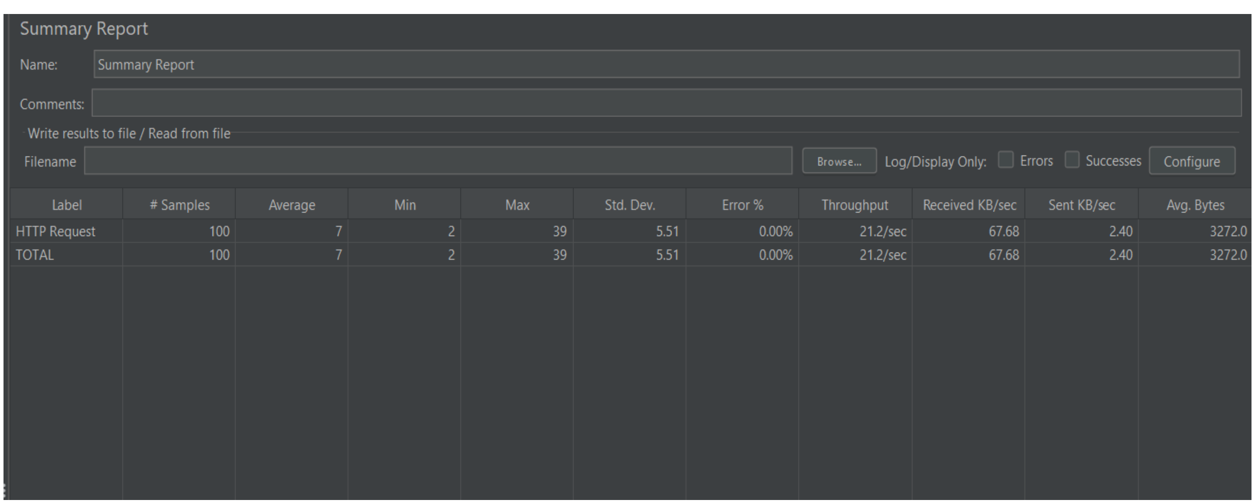
Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Average Response Time	< 2 seconds	7 ms	Pass	Very fast response
2	Maximum Response Time	< 5 seconds	39 ms	Pass	Within acceptable limit
3	Throughput	—	21.2 requests/sec	Pass	Good request handling
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	Normal	Pass	System remained stable
6	Memory Utilization	< 80%	Normal	Pass	Memory usage within limits

Step 4: Observations & Analysis

The SmartLender application performed efficiently during testing. The Random Forest machine learning model required very little processing time because it uses only 11 input features. The average response time remained well below one second, and all test requests were processed successfully without any errors. CPU and memory utilization remained low throughout the testing process, indicating that the application is capable of handling multiple concurrent users in a local deployment environment.

Step 5: Screenshots / Evidence



Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
HTTP Request	100	7	2	39	5.51	0.00%	21.2/sec	67.68	2.40	3272.0
TOTAL	100	7	2	39	5.51	0.00%	21.2/sec	67.68	2.40	3272.0